

ECONOMETRICS I

Problem Set 6

MATÍAS CABELLO
UNIVERSITY HALLE-WITTENBERG

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1. SIMULTANEOUS EQUATIONS: CRIME AND POLICE

Consider the following simultaneous equations model that describes the relationship between crime rates and police deployment:

$$\begin{aligned}\text{Crime} &= \alpha_c + \beta_c \text{Police} + \gamma_c \text{Unemp} + \varepsilon_c \\ \text{Police} &= \alpha_p + \beta_p \text{Crime} + \gamma_p \text{RandBudg} + \varepsilon_p\end{aligned}$$

where:

- **Crime** is the crime rate
- **Police** is police presence
- **Unemp** is the unemployment rate
- **RandBudg** is randomly assigned police budget
- ε_c and ε_p are error terms

The true parameter values are: $\beta_c = -2$ (effect of police on crime) and $\beta_p = 1.5$ (effect of crime on police).

Run the provided R code to generate the data and estimate the model.

- (a) You want to estimate the effect of Police on Crime. Plot Crime vs. Police and explain why ordinary least squares (OLS) regression of Crime on Police would yield biased estimates of β_c .
- (b) Given the equations above, what variable could be used as an instrument to estimate the effect of Police on Crime? Discuss the relevance and exclusion restrictions.
- (c) The manual 2SLS approach involves two stages. Write the equations for both stages and interpret the results from the R output.
- (d) Given the equations above, what variable could be used as an instrument to estimate the effect of Crime on Police? Discuss the relevance and exclusion restrictions.

- (e) Compare the 2SLS estimates with the true parameter values. Why might the estimates not exactly equal the true values?
- (f) Suppose you had data on both RandBudg and Unemp. How would you test whether these instruments are strong enough? What diagnostic would you examine?

2. RETURNS TO EDUCATION: THE COLLEGE DISTANCE NATURAL EXPERIMENT

The `CollegeDistance` dataset of the AER package contains information on wages, education, and proximity to colleges.¹

- (a) Estimate a simple OLS regression of log wages on years of education. Interpret the coefficient and discuss why it might be biased.
- (b) Why can distance to the nearest college serve as a valid instrument for education? Discuss the relevance and exclusion restrictions.
- (c) Estimate the returns to education using college distance as an instrument. Compare the 2SLS estimate with the OLS estimate and interpret the difference.
- (d) Test the strength of the distance instrument. What is the first-stage F-statistic and what does it tell you about instrument relevance?
- (e) Add control variables like ethnicity, gender, and family income to both OLS and IV models. How do the results change and what does this tell us about the identification strategy?
- (f) Some critics argue that college proximity might affect wages through channels other than education (e.g., local labor markets). How could you test this concern empirically?

3. THEORETICAL ASPECTS OF THE COLLEGE DISTANCE STUDY

- (a) In the returns to education literature, why is the ability bias hypothesis typically thought to create upward bias in OLS estimates, while Card's IV estimates are larger than OLS?

¹The data is originally from *Introduction to Econometrics* by Stock and Watson, 2007. Variables (according to the AER package manual):

- **gender** factor indicating gender.
- **ethnicity** factor indicating ethnicity (African-American, Hispanic or other).
- **score** base year composite test score. These are achievement tests given to high school seniors in the sample.
- **fcollege** factor. Is the father a college graduate? **college** factor. Is the mother a college graduate?
- **home** factor. Does the family own their home?
- **urban** factor. Is the school in an urban area?
- **unemp** county unemployment rate in 1980.
- **wage** state hourly wage in manufacturing in 1980.
- **distance** distance from 4-year college (in 10 miles).
- **tuition** average state 4-year college tuition (in 1000 USD).
- **education** number of years of education.
- **income** factor. Is the family income above USD 25,000 per year?
- **region** factor indicating region (West or other).

- (b) How does the college distance instrument relate to the concept of a "natural experiment"? What makes this design more credible than typical observational studies?
- (c) Discuss the external validity concerns of using geographic variation in college proximity to estimate returns to education. For what populations are these estimates most relevant?